Business Requirement Document (BRD)

Project Title: Employee Management & Payroll System

1. Purpose

The purpose of this document is to define the business requirements for developing a centralized

Employee Management & Payroll System using MySQL. This system will manage employee data,

departments, attendance, salary processing, and reporting.

2. Scope

The system will allow HR and Finance teams to manage employee records, track attendance,

process payroll, and generate reports while maintaining data integrity and security.

3. Stakeholders

1 HR Department

2 Finance Team

3 Management

4 IT / Database Administrators

4. Functional Requirements

1 Create, update, and delete employee records

2 Assign employees to departments

3 Track employee attendance

4 Manage salary and payroll details

5 Generate reports for management

5. Non-Functional Requirements

1 Data integrity and consistency

2 High performance and scalability

3 Secure access control

4 Backup and recovery support

6. Assumptions & Constraints

The system will be developed using MySQL 8.x. Users must have basic database knowledge.

Access will be role-based.

7. ER Diagram (Logical View)

Entities: Employees, Departments, Attendance, Salary, Users, Roles

Relationships:

• One Department has many Employees

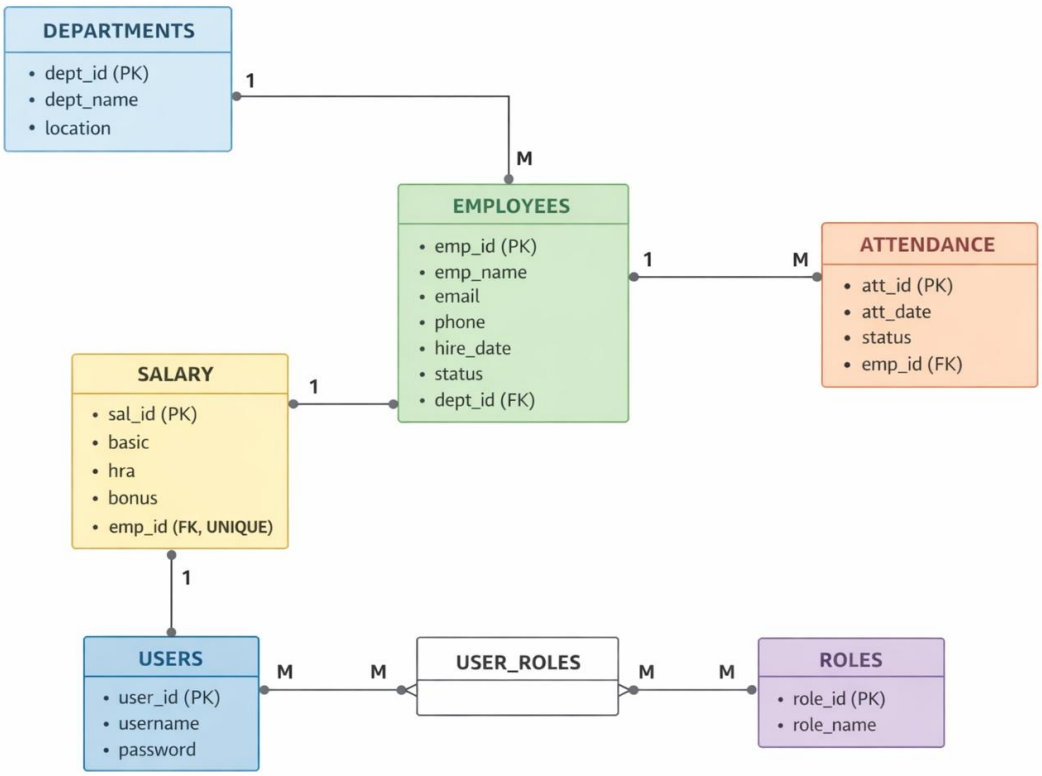
• One Employee has many Attendance records

• One Employee has one Salary record

• Users can have multiple Roles

8. Conclusion

This BRD serves as the foundation for designing and implementing the MySQL-based Employee Management & Payroll System, covering all essential database concepts.



MySQL Programming Questions
(Beginner → Advanced → Real-Time Scenarios)

Beginner Level (DDL + Basic SELECT)

1. Write a query to create the employees table with proper constraints.
2. Insert 5 employee records into the employees table.
3. Display all employees.
4. Display only employee names and email IDs.
5. Find employees whose status is Active.
6. Display employees who joined after 2024-01-01.
7. Rename the column phone to mobile_no.
8. Delete employees whose status is Inactive.

Intermediate Level (WHERE, JOIN, GROUP BY)

9. Display employee name along with department name.
10. Find the number of employees in each department.
11. Display departments having more than 2 employees.
12. Find employees who do not belong to any department.
13. Display employees who do not have salary records.
14. Calculate total salary (basic + hra + bonus) for each employee.
15. Find the highest paid employee.
16. Display employees who were absent at least once.

Subquery & Advanced SELECT

17. Find employees earning more than the average salary.
18. Display employees who earn the maximum salary.
19. Find departments where the average salary is greater than 40,000.
20. Display employees who have never taken leave.
21. Find employees who joined before the oldest attendance record.

Stored Procedures & Functions

22. Write a stored procedure to fetch employees by department ID.
23. Write a stored procedure to insert a new employee.
24. Create a function to calculate total salary.
25. Call the function inside a SELECT query.
26. Write a procedure to update employee status.

Triggers

27. Create a trigger to prevent salary insertion below 10,000.
28. Create a trigger to log deleted employee records.
29. Create a trigger to auto-update status when employee is deleted.
30. Create a trigger to prevent duplicate email insertion.

Transactions (TCL)

31. Write a transaction to insert employee and salary together.
32. Rollback the transaction if salary insert fails.
33. Demonstrate SAVEPOINT usage.
34. Explain what happens if COMMIT is not executed.

Views & Indexes

35. Create a view for HR showing employee & department details.
36. Create a view to show monthly payroll report.
37. Create an index on employee name.
38. Identify which columns should be indexed and why.
39. Drop an index and explain the impact.

Date & Time Queries

40. Find employees who joined in the last 12 months.
41. Calculate employee experience in years.
42. Display attendance count per month.
43. Find employees absent more than 3 days in a month.

Performance & Optimization

44. Use EXPLAIN to analyze a join query.
45. Rewrite a slow query using indexes.
46. Difference between clustered and non-clustered indexes (MySQL context).
47. When should you avoid indexes?

Security & DCL

48. Create a user for HR with read-only access.
49. Grant INSERT permission only on attendance table.
50. Revoke DELETE permission from a user.

Real-Time Scenario-Based (Interview)

51. What happens if department is deleted but employees exist?
52. How will you ensure salary consistency during payroll run?
53. How will you handle partial failure in payroll processing?
54. Explain ACID properties using this project.
55. How will you design audit logging for salary changes?

Coding Challenge (Advanced)

56. Write a single query to display:

- Employee Name
- Department
- Total Salary
- Attendance Count

57. Write a query to rank employees based on salary (without window functions).
58. Find the **top 2 highest paid employees in each department**.